

Q1

```
1  #include<stdio.h>
2  int main()
3  {
4      float P,R,T,SI;
5      printf("ENTER THE VALUE OF PRINICIPAL AMOUNT: ");
6      scanf("%f",&P);
7      printf("ENTER THE VALUE OF RATE OF INTEREST: ");
8      scanf("%f",&R);
9      printf("ENTER THE VALUE OF TIME PERIOD: ");
10     scanf("%f",&T);
11     SI = (P*R*T)/100;
12     printf("THE SIMPLE INTEREST = %.2f\n",SI);
13     return 0;
14 }
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS COD
rams\Lab_1\" ; if ($?) { gcc 1st_file.c -
ENTER THE VALUE OF PRINICIPAL AMOUNT: 100
ENTER THE VALUE OF RATE OF INTEREST: 0.5
ENTER THE VALUE OF TIME PERIOD: 2
THE SIMPLE INTEREST = 1.00
PS C:\Users\RUTVI MAKAWANA\Desktop\VS COD
```

Q2

```
1  #include<stdio.h>
2  int main()
3  {
4      float S1,S2,S3,S4,S5,T,P;
5      printf("ENTER THE MARKS OF SUBJECT-1 OUT OF 100: ");
6      scanf("%f",&S1);
7      printf("ENTER THE MARKS OF SUBJECT-2 OUT OF 100: ");
8      scanf("%f",&S2);
9      printf("ENTER THE MARKS OF SUBJECT-3 OUT OF 100: ");
10     scanf("%f",&S3);
11     printf("ENTER THE MARKS OF SUBJECT-4 OUT OF 100: ");
12     scanf("%f",&S4);
13     printf("ENTER THE MARKS OF SUBJECT-5 OUT OF 100: ");
14     scanf("%f",&S5);
15     T = (S1+S2+S3+S4+S5);
16     P = (T/500)*100;
17     printf("Total marks = %.2f\n",T);
18     printf("Total percentage = %.2f",P);
19     return 0;
20 }
```

output :

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Programs>
($?) { gcc 2nd_file.c -o 2nd_file } ; if ($?) { .\2
ENTER THE MARKS OF SUBJECT-1 OUT OF 100: 89
ENTER THE MARKS OF SUBJECT-2 OUT OF 100: 79
ENTER THE MARKS OF SUBJECT-3 OUT OF 100: 30
ENTER THE MARKS OF SUBJECT-4 OUT OF 100: 70
ENTER THE MARKS OF SUBJECT-5 OUT OF 100: 89
Total marks = 357.00
Total percentage = 71.40
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Programs\
```


Q3

```
#include<stdio.h>
int main()
{
    int B,T,GR,G;
    printf("Enter the basic salary : ");
    scanf("%d",&B);
    printf("Enter the tax paid : ");
    scanf("%d",&T);
    printf("Enter the amount spent on grocery : ");
    scanf("%d",&GR);
    G = (B+T+GR);
    printf("The gross salary is = %d",G);
    return 0;
}
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS COD
($?) { gcc 3rd_file.c -o 3rd_file } ; if
Enter the basic salary : 500000
Enter the tax paid : 100000
Enter the amount spent on grocery : 5800
The gross salary is = 605800
PS C:\Users\RUTVI MAKAWANA\Desktop\VS COD
```

Q4

```
#include<stdio.h>
int main()
{
    float C,F;
    printf("Enter temperature in degree C : ");
    scanf("%f",&C);
    F = (1.8*C)+32;
    printf("The temperature is degree fahrenheit : %f",F);
    return 0;
}
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Programs> gcc 4th_file.c -o 4th_file ; if ($?) { gcc 4th_file.c -o 4th_file } ; if ($?) { gcc 4th_file.c -o 4th_file }
Enter temperature in degree C : 100
The temperature is degree fahrenheit : 212.000000
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Programs>
```


Q5

```
> C 5th_file.c > main()
#include<stdio.h>
int main()
{
    int a,b,c;
    printf("Enter two values for a and b : ");
    scanf("%d%d",&a,&b);
    c=a;
    a=b;
    b=c;
    printf("Now the value of a: %d and\n b: %d",a,b);
    return 0;
}
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\
($?) { gcc 5th_file.c -o 5th_file
Enter two values for a and b : 2
4
Now the value of a: 4 and
b: 2
PS C:\Users\RUTVI MAKAWANA\Desktop\
```

Q6

```
#include<stdio.h>
int main()
{
    int a,b;
    printf("Enter two values of a and b: ");
    scanf("%d%d",&a,&b);
    a = a+b;
    b = a-b;
    a = a-b;
    printf("Now the value a is : %d and b is : %d",a,b);
    return 0;
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Pr
($?) { gcc 6th_file.c -o 6th_file } ; if ($?
Enter two values of a and b: 10
20
Now the value a is : 20 and b is : 10
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Pr
```


Q7

```
#include<stdio.h>
int main()
{
    float B,H,A;
    printf("The base of right angle triangle(in m): ");
    scanf("%f",&B);
    printf("The height of right angle triangle(in m): ");
    scanf("%f",&H);
    A = 0.5*B*H;
    printf("The area of triangle is %f meters sq.",A);
    return 0;
}
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE F
($?) { gcc 7th_file.c -o 7th_file } ; if ($?) {
The base of right angle triangle(in m): 2
The height of right angle triangle(in m): 4
The area of triangle is 4.000000 meters sq.
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE F
```

```

1  #include<stdio.h>
2  int main()
3  {
4      int s=31558150;
5      int D,H,M,RS;
6      D = s/86400;
7      RS = s%86400;
8      H = RS/3600;
9      RS = RS%3600;
10     M = RS/60;
11     printf("Days = %d\n",D);
12     printf("Hours = %d\n",H);
13     printf("Minutes = %d\n",M);
14     return 0;
15
16 }
```

Q8

output :

```

PS C:\Users\RUTVI MAKAWANA\Desktop\VS
($?) { gcc 8th_file.c -o 8th_file } ;
Days = 365
Hours = 6
Minutes = 9
PS C:\Users\RUTVI MAKAWANA\Desktop\VS
```


Q9

```
#include<stdio.h>
int main()
{
    int H,M,S,TS;
    printf("Enter the time in hours: ");
    scanf("%d",&H);
    printf("Enter the time in minutes: ");
    scanf("%d",&M);
    printf("Enter the time in seconds: ");
    scanf("%d",&S);
    TS = ((H*3600)+(M*60)+S);
    printf("The total time in seconds is : %d",TS);
    return 0;
}
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE
($?) { gcc 9th_file.c -o 9th_file } ; if
Enter the time in hours: 4
Enter the time in minutes: 10
Enter the time in seconds: 5
The total time in seconds is : 15005
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE
```

Q10

```
1  #include<stdio.h>
2  void main()
3  {
4      float CM,M,P,C,E;
5      printf("Enter the marks of mathematics out of 200 : ");
6      scanf("%f",&M);
7      printf("Enter the marks of Physics out of 200 : ");
8      scanf("%f",&P);
9      printf("Enter the marks of Chemistry out of 200 : ");
10     scanf("%f",&C);
11     printf("Enter the marks of Entrance exam out of 100 : ");
12     scanf("%f",&E);
13     CM = (M/2+P/2+C/2+E);
14     if(CM>=500)
15     {
16         printf("You have Cleared the Cutoff!!");
17     }
18     else
19     {
20         printf("Oops! you missed the cutoff");
21     }
22 }
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Program
($?) { gcc 10th_file.c -o 10th_file } ; if ($?) {
Enter the marks of mathematics out of 200 : 189
Enter the marks of Physics out of 200 : 178
Enter the marks of Chemistry out of 200 : 160
Enter the marks of Entrance exam out of 100 : 70
Oops! you missed the cutoff
PS C:\Users\RUTVI MAKAWANA\Desktop\VS CODE Program
```


Q11

```
1  #include<stdio.h>
2  int main()
3  {
4      int s,D,H,M,RS;
5      printf("Enter the seconds: ");
6      scanf("%d",&s);
7      D = s/86400;
8      RS = s%86400;
9      H = RS/3600;
10     RS = RS%3600;
11     M = RS/60;
12     printf("Days = %d\n",D);
13     printf("Hours = %d\n",H);
14     printf("Minutes = %d\n",M);
15     return 0;
16
17 }
```

output:

```
PS C:\Users\RUTVI MAKAWANA\Desktop
($?) { gcc 11th_file.c -o 11th_f
Enter the seconds: 15005
Days = 0
Hours = 4
Minutes = 10
PS C:\Users\RUTVI MAKAWANA\Desktop
```